TRIBHUVAN UNIVERSITY

INSTITUTE OF ENGINEERING

PASHCHIMANCHAL CAMPUS

A MID-TERM REPORT

ON

DESIGN AND FABRICATION OF ELECTRIC LAWN MOWER WITH PERFORMANCE ANALYSIS OF DIFFERENT CUTTING BLADES

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ABSTRACT

The goal of this project is to design and fabricate a cost-effective battery-powered rotary lawn mower with interchangeable cutting blades and a user-friendly height adjustment mechanism. Electric mowers offer clear advantages such as lower noise, zero emissions, and reduced running costs, but blade performance across varying turf conditions in Nepal remains under-researched. This project aims to bridge that gap by developing a versatile mower system that ensures efficient grass cutting while minimizing power consumption.

The mower operates on a 24 V DC supply and incorporates a PWM-based motor speed controller tuned according to blade geometry. Blades will be selected according to their market availability and rated performance, and their efficiency, durability, and power requirements will be evaluated under different grass species and conditions.

CAD tools will be employed for blade and chassis modeling, supported by Finite Element Analysis (FEA) for stress and deformation testing. The blade-lock and height-adjustment modules will be refined through rapid prototyping and user trials, while field experiments will assess cutting uniformity, energy efficiency, and ergonomics.

Expected outcomes include optimized blade profiles for consistent performance, an embedded speed-control algorithm for energy efficiency, and a practical tool-free height and blade interchange system. The mower is envisioned as a sustainable, user-friendly solution for small-scale landscapers and homeowners.

Keywords: Cutting Blades, Electric Lawn mower, Power consumption, Speed Controller

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CHAPTER ONE: INTRODUCTION

An electric lawn mower operates on the principle of converting electrical energy into mechanical energy to facilitate grass cutting. At its core, an electric motor receives power, either from a rechargeable battery or a direct connection to a standard electric outlet via a power cord. This electrical input is then transformed into rotational motion, which drives the mower's blades. These blades, typically crafted from durable materials such as low-carbon alloy steel, are engineered to maintain sharpness and rotate horizontally, ensuring an efficient and even cut across the lawn. User control is facilitated through intuitive mechanisms, including on/off switches, variable speed settings that adjust blade rotation, and adjustable height settings for the cutting deck, allowing for precise grass length management.

Electric lawn mowers have gained increasing prominence in recent years as a cleaner, quieter, and more sustainable alternative to traditional gas-powered models. This shift is being driven by a combination of environmental, technological, and policy-related factors that are converging to reshape the lawn care industry's landscape. Historically, lawn maintenance has relied heavily on small internal combustion engines, which contribute significantly to local air pollution and greenhouse gas emissions. With growing awareness of climate change and air quality concerns, this has created substantial public and regulatory pressure to transition to electric solutions.

Figure 1: Electric Lawn Mower

1.1 Background

The idea of the electric lawn mower dates back to the late 1920s and 1930s, when the first corded models appeared. These early machines provided a cleaner alternative to gasoline engines but were limited by their tether to a power outlet, making them suitable only for small lawns and requiring constant cord management. By the 1950s, improvements in electric motor technology made them more powerful and efficient, expanding their practicality. A major breakthrough came in the 1980s with cordless mowers powered by lead-acid batteries, which allowed greater mobility. However, these batteries were heavy, short-lived, and typically lasted only 2–4 years even with proper care.

The real revolution began in the 2000s with the adoption of lithium-ion batteries. Offering nearly five times the energy density of lead-acid, they provided longer run-times, higher efficiency (above 90%), and a much longer service life 5–10 years or 500–800 charging cycles, while recharging in just 1–2 hours. Recent advances have introduced Lithium Iron Phosphate (LiFePO4) batteries, a superior variant of lithium-ion, delivering lifespans beyond 10 years or up to 3,000 cycles. They are also safer and perform reliably across a wider temperature range, from –20°C to 75°C.

Motor technology has also advanced significantly. Early mowers relied on brushed motors, but modern designs increasingly use brushless motors controlled electronically. By eliminating brushes, these motors reduce friction and energy losses, producing less heat and greater efficiency. This means more power reaches the blades, allowing tougher grass to be cut with ease. Brushless motors also run quieter, require less maintenance, and last much longer—often exceeding 10,000 hours of use.

1.2 Statement of Problem

Following are the existing problems concerns in the project:

i. Same cutting blade is not suitable for different types of grass field.
   - Finer grasses may require a sharper, lightweight blade for clean cutting, while thicker or tougher grasses demand a sturdier profile to avoid clogging or uneven cuts. Using the same blade everywhere often leads to poor efficiency and unsatisfactory results.

ii. Different cutting blades have different amount of power consumption rate.
   - Some blades require more torque to cut effectively, which increases the overall energy consumption and reduces battery runtime. This makes it difficult to achieve both clean cutting and energy efficiency with a single type of blade.

1.3 Scope of Work

This project involves the design and fabrication of electric lawn mower with performance analysis of different cutting blades in the terms of power consumption and cutting efficiency. This project will have concerns on the speed controllability, interchangeability of cutting blades and height adjustment of deck or housing. The project will also involve CAD modeling, basic structural analysis, and the construction of a working prototype. Lawn mower will be tested in different lawn fields having different grass density.

However, Advance automation, durability testing, regulatory certification and mass production will be beyond the concerns of the project. Also, the unavailability of ideal grass field could be major challenges of the project.

1.4 Objectives

The objectives of the project are divided into main objectives and specific objectives, which are mentioned below:

Main Objective:

i. To design and fabricate an electric lawn mower and performance analysis of different cutting blades.

Specific objectives:

i. To design and fabricate an electric lawn mower.

ii. To introduce a speed controller for cutting blades.

iii. To optimize the height adjustment system on the lawn mower.

iv. To analyze the cutting efficiency and power consumption of different types of cutting blades.

CHAPTER TWO: LITERATURE REVIEW

Ziauddin (2011) developed a small-scale electric mower and emphasized how grass type and density significantly influenced cutting quality and power draw. The work included controlled experiments on varying field conditions, showing that efficiency dropped sharply when a single blade profile was used across multiple species. This highlighted the practical challenge of designing a universal blade system and laid the groundwork for adaptive blade geometry research.

Okafor (2013) presented a design for a self-powered mower that integrated collapsible blades with a pulley-driven DC motor system. Beyond durability, the study stressed cost-effectiveness and portability, particularly in rural environments where affordability and ease of maintenance are critical. The inclusion of an alternator to recharge the battery during operation demonstrated early innovation toward energy self-sufficiency in mowing systems.

Rahul et al. (2015) performed detailed CAD modeling and Finite Element Analysis (FEA) on rotary mower blades to evaluate stress distribution and fatigue life. Their ANSYS-based simulations confirmed that bevel angles of 25°–30° not only reduced torque demand but also improved structural reliability under repeated load cycles. The findings also suggested that improper blade geometry accelerates wear, causing higher power consumption and reduced battery runtime, making angle optimization essential.

Nagarajan et al. (2017) focused on fabrication and testing of a mower with helix-shaped blades. The research prioritized lightweight materials and a compact frame to enhance maneuverability. Performance testing revealed that the helix design provided smoother cutting action, especially on uneven terrain, compared to flat blades. The authors concluded that unconventional geometries could improve efficiency for household and small-scale agricultural use.

Akene et al. (2020) developed and tested a solar-powered mower with adjustable wheel heights, assessing its performance across different grass types. Their evaluation revealed that efficiency ranged between 70–82% depending on species and height setting, confirming the strong influence of adjustment mechanisms on performance. The project

also reinforced sustainability by demonstrating the viability of renewable energy integration in lawn care equipment.

Messaadi and Amroun (2021) applied a fuzzy-PID controller to DC motors and compared it with conventional PID systems. The fuzzy-PID design showed faster settling time, reduced overshoot, and more stable operation under load variations. By mitigating sudden torque and current spikes, the controller improved both energy efficiency and motor protection, making it a promising solution for battery-operated lawn mowers where power management is crucial.

Li and Chen (2024) contributed a theoretical and experimental study on straight-blade roller mower design. They found that sliding-cutting angles between 25° and 30° combined with a grind-edge radius of approximately 0.05 mm significantly reduced chopping power across turf species. Their results offered practical, data-backed guidelines for designing blades that minimize energy use while maintaining cutting efficiency, reinforcing the importance of blade geometry optimization for sustainable mower design.

CHAPTER THREE: METHODOLOGY

Methodology to achieve the project objectives, the following steps will be undertaken

i. Literature Review:

Conduct a review of existing electric rotary lawn mowers to identify current designs, technologies, and limitations. The study will cover motor and battery options, PWM-based speed control, and blade geometries suited for different grass types. Ergonomics, safety standards, and ease of height/blade adjustment will also be evaluated. Finally, cost analysis of key components will help identify opportunities for affordability and design optimization.

ii. Conceptual Design:

Develop a set of technical specifications based on available technology. Create initial design sketches and concepts that reflect cost-effective choices for motors, batteries, controller, and structural materials. This phase will focus on optimizing the balance between affordability, durability, and functionality.

iii. Detailed Design and Modeling:

Use Computer-Aided Design (CAD) software to create detailed 3D models of the lawnmower, ensuring that critical factors such as weight distribution, stability, and material strength are considered. Conduct simulations and Finite Element Analysis (FEA) to evaluate the lawnmower's performance under real-world conditions, particularly focusing on force distribution, stability, and durability.

iv. Fabrication:

Source materials that align with the design specifications. Fabrication will involve processes like cutting, machining, welding and fastening to assemble a sturdy yet lightweight frame. Electrical components such as brushless DC motor, battery packs, and PWM controller will be integrated to complete the assembly.

v. Model development and Testing:

Build a fully functional model of the lawnmower. Testing will be conducted to evaluate the functionality of the height-adjustment system, motor speed control, maneuverability, blade interchangeability and battery life. Tests will be performed in both controlled

environments and real-world conditions to ensure the lawnmower's performance across various terrains.

vi. Trials and Data Gathering:

Conduct trials in real environment to gather data on the lawnmower's performance, ease of use, and overall efficiency. This will be invaluable in identifying areas for improvement, ensuring that the final product is not only functional but also meets the main as well as specific objectives.

vii. Performance Evaluation and Validation:

Analyze the results of the trials and technical tests to validate the lawnmower's performance, safety, and reliability. Compare the findings against established benchmarks for electric lawnmowers and make necessary adjustments to optimize the design.

viii. Documentation and Reporting:

Prepare detailed documentation outlining the entire design, development and testing process. The final report will summarize research findings, design considerations, fabrication steps, testing outcomes, and cost analysis, ensuring a comprehensive overview of the project.

This methodology ensures that a fully functioning electric lawnmower model is developed, meeting both technical and objective-centric requirements, while remaining affordable and accessible.

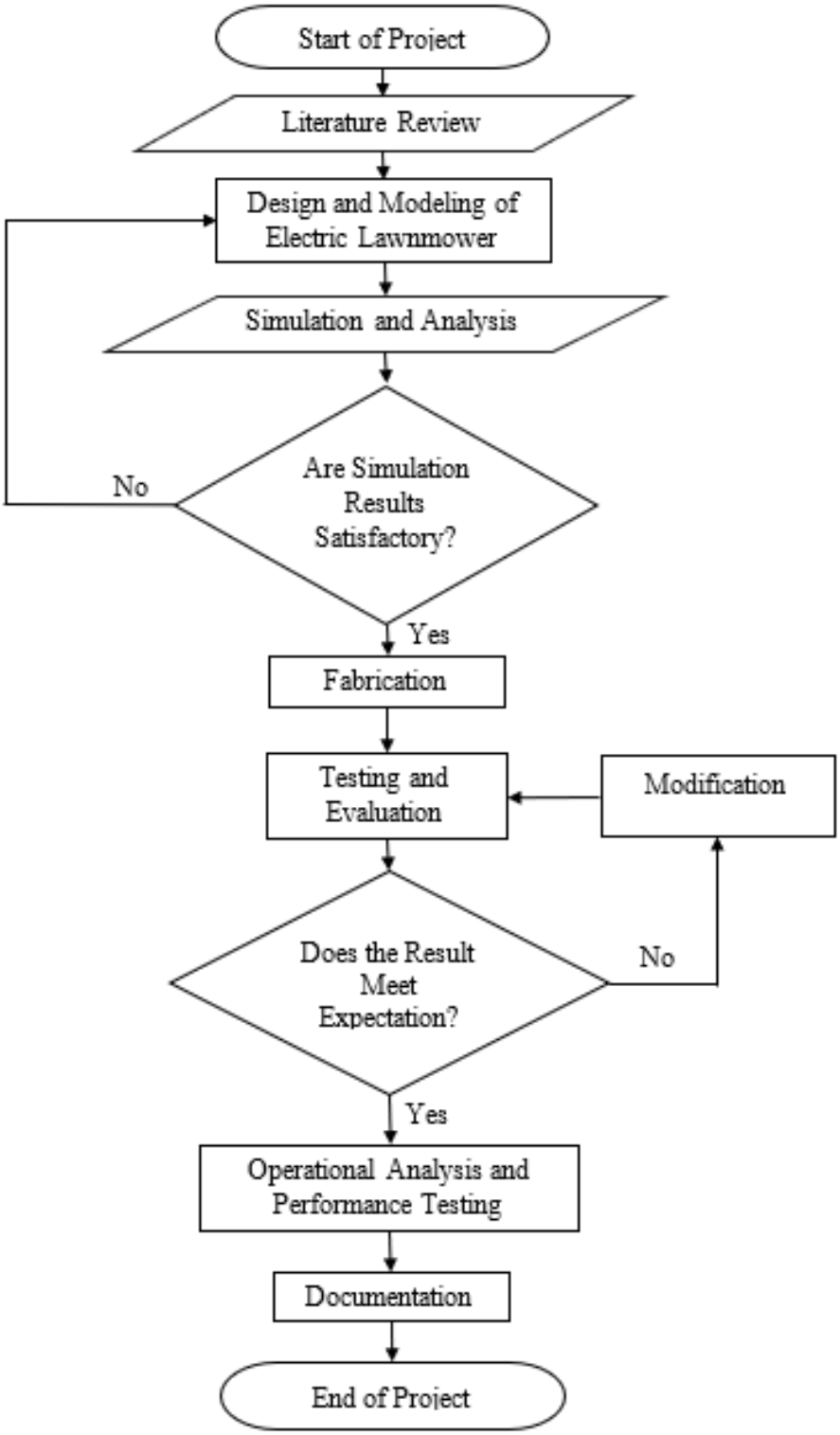

Figure 2: Flowchart of Methodology

CHAPTER FOUR: WORK ACCOMPLISHED

PHASE I: Literature Review

In the early phase of the project, our team focused on conducting a detailed literature review to build a solid foundation for the design of the electric lawnmower. The review concentrated on several important aspects, including blade geometry, grass-cutting efficiency, motor power requirements, battery capacity, handling ergonomics, and safety features. Considerable attention was also given to maneuverability, ensuring that the mower can operate effectively under varying grass and field conditions. This stage helped us identify both the strengths and shortcomings of existing designs, which in turn guided us toward solutions better suited to our objectives.

To strengthen the study, the team carefully selected a broad range of technical papers, design standards, and academic journals as key reference materials. By critically reviewing and summarizing these sources, we gained valuable insight into both the practical challenges and theoretical principles underlying electrically driven machines. This extended to examining areas such as battery technologies, electric motor performance, dynamic stability, user–machine interface, and assistive control systems. The knowledge gained from this process not only deepened our understanding of the engineering considerations but also shaped many of the design choices that followed, ensuring the development of a more reliable, efficient, and user-oriented electric lawnmower.

PHASE II: Design Consideration

A. Measurement of Chassis Dimensions:

In the first stage of design consideration, the project team started by measuring the geometry of market available cutting blades, and designed the dimensions of Motor Base Plate, frame and all other parts as per allowing cutting blade to operate with proper clearance.

Table 4.1: Electric Lawn Mower Specifications

Specification	Dimensions (mm)
Cutting Blade (Max.)	380
Motor Base Plate and Blade Housing	480
Frame Width	540
Frame length	760
Handle Height	850
Blade Height Range	20-90

The project team examined the dimensions of various lawnmower models to understand the standard size requirements. Based on the specified measurements and the project's design requirements, the frame for the new electric lawnmower was designed with a length of about 760 millimeters and a width of approximately 540 millimeters.

B. Selection of Material:

For the fabrication of the electric lawn mower, materials were evaluated considering strength, cost, and outdoor suitability. Mild steel was selected for the frame as it offers adequate strength, rigidity, and ease of fabrication at a reasonable cost, making it practical for withstanding operational loads while keeping manufacturing simple. Cast iron was chosen for the motor deck and height-adjusting mechanism because of its high compressive strength, wear resistance, and vibration-damping capacity. These properties ensure stable motor mounting, smoother operation, and reliable performance of the adjustment system under continuous stress. The combination of mild steel and cast iron provides the necessary structural integrity, durability, and affordability, making it a balanced and effective choice for long-term use of the mower.

C. Selection of Cross-Section:

After deciding on mild steel as the most suitable material for the electric lawn mower chassis, the next step was to determine the best cross-section for the frame.

From analysis of existing designs of lawn mowers, it was clear that square tubing generally provides greater strength and durability than other kinds of tubing. Square sections distribute stress more evenly and resist twisting under load. This makes them particularly well-suited for a lawn mower chassis, where vibration and uneven terrain are expected.

To validate this finding, the project team square pipe sections with nearly identical dimensions in CAD software. The design was then tested under the same simulated loading conditions. The results reinforced the theoretical analysis, confirming that the square performed well in deformation testing under our expected loading conditions.

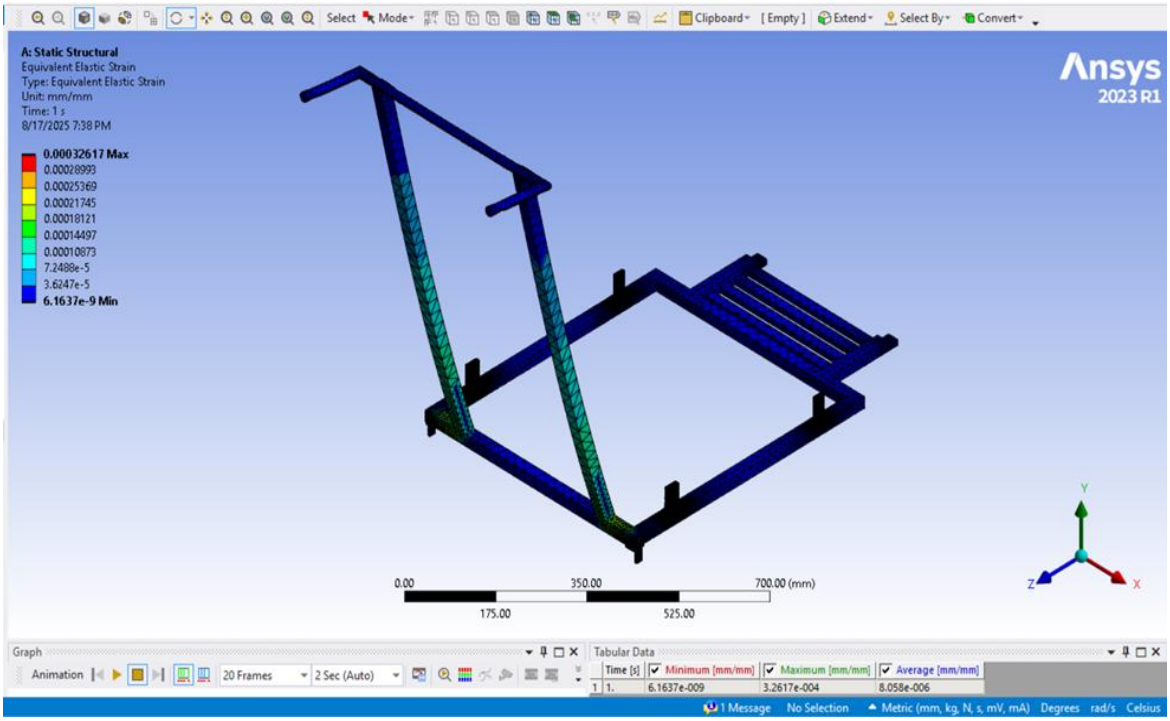

Figure 3: Deformation Test simulation performed on square tube

D. Selection of Electric Motor:

As part of the electric lawn mower design project, the team faced the challenge of selecting an appropriate motor to ensure reliable cutting performance, energy efficiency, and ease of operation. A thorough investigation was conducted into suitable motor types, with careful consideration of factors such as power output, weight, torque, and efficiency.

The literature review and market research highlighted that brushless motors are widely applied in compact outdoor equipment due to their durability, smooth operation, and favorable balance between cost and performance. The team also evaluated user requirements including consistent cutting power, battery compatibility, and adaptability to varying grass and terrain conditions.

After assessing multiple alternatives, the team selected a motor with a power rating of 600-700 watts. The specification provides an optimal balance between torque and rotational speed, ensuring effective grass cutting while maximizing battery life. The chosen motor supports the project's objectives of building a dependable, user-friendly, and energy-efficient electric lawn mower.

E. Selection of Battery

In selecting a suitable battery for the electric lawn mower, lithium-ion batteries are preferred over traditional lead-acid options due to their significant advantages. Lithium-ion batteries are lighter, have a longer service life, and support a higher number of charge cycles. They also recharge more quickly, making them well-suited for repeated lawn care tasks. Lithium-ion batteries are deep-cycle, meaning they can handle frequent charging and discharging without much loss in performance.

By contrast, lead-acid batteries are heavier, bulkier, have a shorter lifespan, and require longer charging times, though they are less expensive initially. For a 24V electric lawn mower system, typically achieved by connecting two 12V batteries in series, a 40Ah lithium-ion battery offers a practical balance of cutting time, efficiency, and reliability. This choice aligns with the project's goals of creating a lightweight, durable, and user-friendly electric mower.

PHASE III: Feasibility Study

This study looks at the practicality of designing and building a working prototype of an electric lawn mower, with a focus on technical needs, material choices, and how the design can be put into action. The project team will carry out the full process of fabrication and assembly themselves. Based on the analysis, the project is considered technically achievable, for several reasons:

i. The design goals and specifications for the electric lawn mower have already been clearly outlined. This gives the team a solid roadmap to follow, keeping the build process organized and aligned with the project's objectives.

ii. The team has the right engineering knowledge and hands-on skills to take care of the design, construction, and assembly. Their experience will help ensure the mower is not only reliable and efficient, but also practical and easy to use.

iii. The team is dedicated to completing a fully operational prototype. A testing phase is built into the plan to confirm that the mower performs well under real-world grass-cutting conditions. The group has also anticipated possible challenges in fabrication and assembly, with backup strategies in place to keep the process on track.

Taking all of this into account, the project of building a fully functional electric lawn mower is realistic and achievable. With a clear plan, access to the right components, and the team's expertise, the project stands on strong ground for successful completion.

A. Performance parameters

As part of the feasibility study, various performance calculations were carried out to assess the practicality of the electric lawn mower design. These included evaluating the required torque for effective grass cutting, the maximum operating speed, the mower's ability to handle different lawn conditions and inclines, and the estimated operating range on a single battery charge.

Assumptions and Standard values

Shearing force for most lawn grasses is (9.2-11.5 N) [Okafor, 2013], We assume 10.5N,

Blade Geometry **(**From field and FE studies):

Cutting angle: 25-30° (30-40° acceptable),

Cutting edge radius: 0.05mm,

Motor Power and Shaft Torque:

Widely cited mower design guidance recommends, Motor Power ≥ 628w

Blade speed = 2000-3000 RPM

For High Lift Cutting Blade:

Length (l) = 380mm,

Thickness (t) = 2mm,

Width (w) = 38mm,

Weight (W) = 600gm,

Force required by cutting blade to shear grass,

F = T/R (Khurmi, 2003)

Torque (T) = 10.5 * 0.19 = 1.995 Nm

To produce 'T' of 1.995 Nm,

For a 3000rpm motor,

Power (P)= $2\pi NT/60$ = 627 W

For Disc Blade:

Diameter (D) = 255mm,

Inner Diameter (d) = 25.4mm,

Thickness (t) = 2mm,

F=T/R

Torque (T) = 10.5 * 127.5 = 1.333875 Nm

Power (P) = $2\pi NT/60$ = 420w

For Straight Blade (Dimensions Same as High Lift Cutting Blade)

Length (l) = 380mm,

Thickness (t) = 2mm,

Width (w) = 38mm,

Weight (W) = 600gm,

Force required by cutting blade to shear grass,

F = T/R (Khurmi, 2003)

Torque (T) = 10.5 * 0.19 = 1.995 Nm

To produce 'T' of 1.995 Nm,

For a 3000rpm motor,

Power (P)= $2\pi NT/60$ = 627 W (We intend to use 650W)

Battery Sizing:

Efficiency (η) = 0.85 (Standard Assumptions)

Battery Capacity = Power (P) * Operating time / Voltage * DOD * Efficiency

Where,

Depth of discharge (DOD) = 0.8% for Li- ion batteries,

Power (P) = 650w,

Voltage (V) = 24v,

Operating Time = 1hr,

Battery Capacity = 650*1/ 24*0.8*0.85 = 37 Ah or 35-40Ah

PHASE IV: Design of The Project

The design process for the electric lawn mower was carried out using SolidWorks software, which allowed the team to visualize and create each component before fabrication. Every part was modeled with care to ensure the mower would be efficient, stable, and easy to operate. The key components designed include:

I. **Frame:** The main structure of the mower, built to provide strength and stability while securely supporting the motor, battery, and cutting system.

II. **Motor Mounts:** Specially designed supports that hold the electric motor in place, ensuring smooth power transfer to the blade and wheels.

III. **Wheel and Support System:** Developed to provide steady movement across different lawn conditions, ensuring durability, balance, and ease of maneuverability during operation.

Figure 4: CAD Model of Electric Lawn Mower

CHAPTER FIVE: PROJECT TIMELINE

The estimated timeline of this project is shown as in table 5.1. The project would take thirty-six weeks to complete. At the end of thirty-fourth week tested model would be ready with final report.

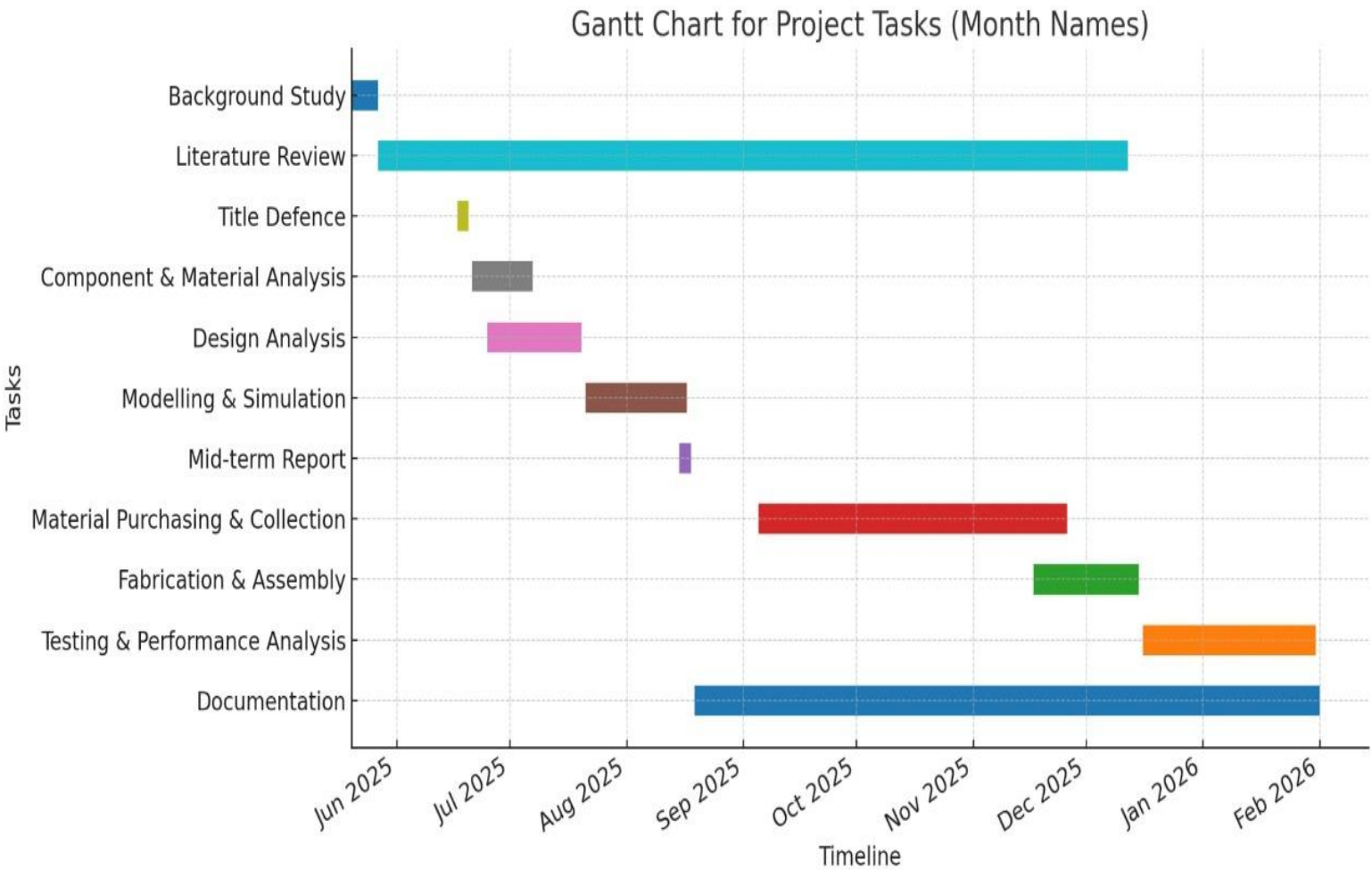

Figure 5: Gantt Chart

CHAPTER SIX: CONCLUSION AND WORK REMAINING

6.1 Conclusion

In conclusion, the development of electric lawn mower has shown promising progress, addressing key challenges in creating a machine that is both efficient and user-friendly. Research and market analysis provided valuable insights into user preferences, safety considerations, and performance needs, helping guide the design toward practicality and comfort. Careful attention was given to selecting cost-effective motors, batteries, and structural materials to strike the right balance between durability, performance, and affordability. The team translated these ideas into detailed 3D models using CAD tools, supported by simulations and Finite Element Analysis (FEA) to ensure reliability, strength, and safe operation under real conditions. Ultimately, this project has the potential to deliver a sustainable and accessible lawn care solution, reducing the physical effort required while promoting environmentally friendly practices. By investing in this innovation, the project takes an important step toward smarter, greener, and more convenient outdoor maintenance tools.

6.2 Work Remaining

The remaining objectives that need to be achieved for the electric lawn mower project are as follows:

i. Further literature review is to be done.
ii. Purchasing and Collection of material is to be done.
iii. Fabrication of the fully functioning model is to be done.
iv. Testing and fine tuning of the model is to be done.
v. Documentation and reporting of project is to be done.

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